gpu-fryer summary

- Generated: 2026-08-06 17:19:05
- Nodes: node5500 (8 x B200), node5501 (8 x B200), node5502 (8 x B200)
- Note: node5500-5502 are no longer in Slurm. The tables below are their historical baseline; the current hardware is the 7 new nodes covered in "New nodes" below.
- Precisions: FP32, BF16, FP8
- Reference (MIT aicr-benchmarks, `gpu-fryer/summary.md`, b0025, **B200**): per-GPU mean TFLOP/s — FP32 772, BF16 1500, FP8 4115

Per-node mean converged throughput (TFLOP/s)

Node	GPU	FP32	BF16	FP8	Health
node5500	B200	744	1457	3990	ok
node5501	B200	748	1457	4001	ok
node5502	B200	760	1437	4062	ok
reference (b0025)	**B200**	**772**	**1500**	**4115**	—

% of B200 reference (mean, B200 nodes only)

Node	FP32	BF16	FP8
node5500	96%	97%	97%
node5501	97%	97%	97%
node5502	98%	96%	99%

New nodes (node5600/5601/5602/5702/5800/5801/5802)

Run 2026-08-24 on the 7 new B200 nodes, same configuration as above (300 s per precision, all 8 GPUs, 7 single-node jobs).

They match the nodes above — no separate tables needed. Per-node means land at FP32 740-752, BF16 1451-1465, FP8 3949-4032 TFLOP/s. Against the node5500-5502 means the 7-node means differ by **-0.4% (FP32), +0.5% (BF16), -0.3% (FP8)**, and no individual node is more than **1.7%** away on any precision (worst: node5600-c1, FP8). Every node lands at 96-98% of the b0025 B200 reference, the same band the old nodes occupied.

All 7 are healthy. gpu-fryer reports "All GPUs seem healthy" on every node;

no HW, thermal-SW or thermal-HW throttling flag was raised on any of the 56 GPUs.

Per-GPU uniformity is also unchanged: at 8 GPUs the derived speed-up is

7.81-7.88x, against 7.87-8.03x on the old nodes.

Speed-up vs number of GPUs

`![Speed-up vs number of GPUs](gpu-fryer-speedup.svg)`

Shown for **node5500**, one curve per precision.

#GPUs	BF16	FP32	FP8	ideal
1	1.00	1.00	1.00	1.00
2	1.99	2.02	2.00	2.00
3	2.97	3.02	2.97	3.00
4	3.94	4.01	3.95	4.00
5	4.93	5.03	4.93	5.00
6	5.92	6.03	5.93	6.00
7	6.90	7.03	6.90	7.00
8	7.87	8.03	7.88	8.00

The other nodes (node5501, node5502) are close and are omitted from the plot to keep it readable: at 8 GPUs node5500 reaches BF16 7.87x, FP32 8.03x, FP8 7.88x, and no other node differs by more than **0.16x** on any precision.

> **How to read this.** gpu-fryer stresses all 8 GPUs *concurrently* and reports one converged figure per GPU — it does not run separate 1, 2, ... 8-GPU jobs. The curve above is therefore **derived** from that single run: speed-up(N) = (sum of GPUs 0..N-1) / GPU 0. It is linear by construction and is **not** a measured scaling study; what it shows is per-GPU *uniformity* — a curve that tracks the dashed ideal line means every GPU sustains the same throughput, while a curve bending below it marks a slow or throttling GPU. For real scaling behaviour see the Megatron-LM weak-scaling results in `output-megatron/summary.md`.

Per-GPU converged throughput (TFLOP/s)

node5500 (8 x B200)

GPU	FP32	BF16	FP8
0	740.9	1480.4	4049.2
1	755.1	1470.5	4064.6
2	738.5	1440.6	3921.7

3	736.1	1439.6	3940.8
4	752.7	1470.3	4002.6
5	747.9	1458.6	4035.6
6	740.8	1450.2	3920.9
7	740.9	1444.2	3988.4
min	**736.1**	**1439.6**	**3920.9**
mean	**744.1**	**1456.8**	**3990.5**
max	**755.1**	**1480.4**	**4064.6**

node5501 (8 x B200)

GPU	FP32	BF16	FP8
0	759.2	1476.6	4059.5
1	747.3	1452.8	4012.1
2	742.4	1446.1	3993.2
3	754.4	1469.3	4026.3
4	735.4	1429.2	3974.8
5	749.6	1460.4	4045.5
6	751.9	1471.7	3936.2
7	740.0	1446.1	3960.1
min	**735.4**	**1429.2**	**3936.2**
mean	**747.5**	**1456.5**	**4001.0**
max	**759.2**	**1476.6**	**4059.5**

node5502 (8 x B200)

GPU	FP32	BF16	FP8
0	759.7	1445.1	4087.0
1	769.3	1441.1	4113.7
2	755.0	1434.7	4048.0
3	764.5	1447.6	4092.0
4	766.8	1444.8	4063.4

5	757.4	1432.0	4011.1
6	755.1	1424.9	4058.6
7	754.5	1425.6	4026.0
min	**754.5**	**1424.9**	**4011.1**
mean	**760.3**	**1437.0**	**4062.5**
max	**769.3**	**1447.6**	**4113.7**

Converged = the final sustained-average throughput gpu-fryer reports per GPU at the end of each precision run. Higher is better; large spread across GPUs or any throttling flag indicates a problem.